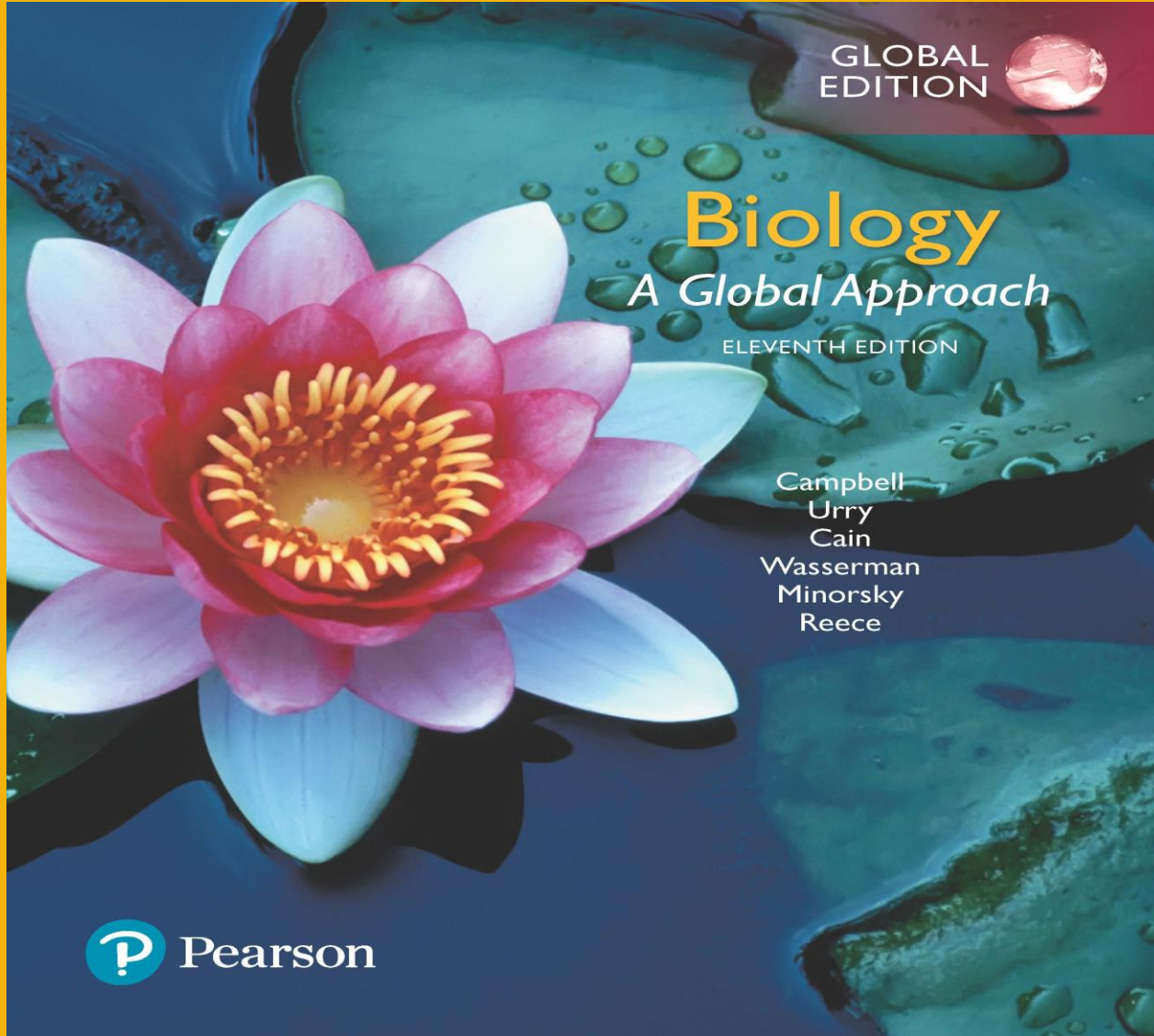




Chapter 26

Introduction to Viruses

Lecture Presentations by
Nicole Tunbridge and
Kathleen Fitzpatrick

A Borrowed Life

- A **virus** is an infectious particle consisting of genes packaged in a protein coat
- Viruses are much simpler in structure than even prokaryotic cells
- Viruses cannot reproduce or carry out metabolism outside of a host cell
- Viruses exist in a shady area between life-forms and chemicals, leading a kind of “borrowed life”

Concept 26.1: A virus consists of a nucleic acid surrounded by a protein coat

- Viruses were detected indirectly long before they were actually seen

The Discovery of Viruses: *Scientific Inquiry*

- Tobacco mosaic disease stunts growth of tobacco plants and gives their leaves a mosaic coloration
- In the late 1800s, researchers hypothesized that unusually small bacteria might be responsible
- Later work suggested that the infectious agent did not share features with bacteria (such as the ability to grow on nutrient media)
- In 1935, Wendell Stanley confirmed this latter hypothesis by crystallizing the infectious particle, now known as tobacco mosaic virus (TMV)

Figure 26.2

Experiment



1 Extracted sap from tobacco plant with tobacco mosaic disease



2 Passed sap through a porcelain filter known to trap bacteria



3 Rubbed filtered sap on healthy tobacco plants



4 Healthy plants became infected

Structure of Viruses

- Viruses are not cells
- A virus is a very small infectious particle consisting of nucleic acid enclosed in a protein coat and, in some cases, a membranous envelope

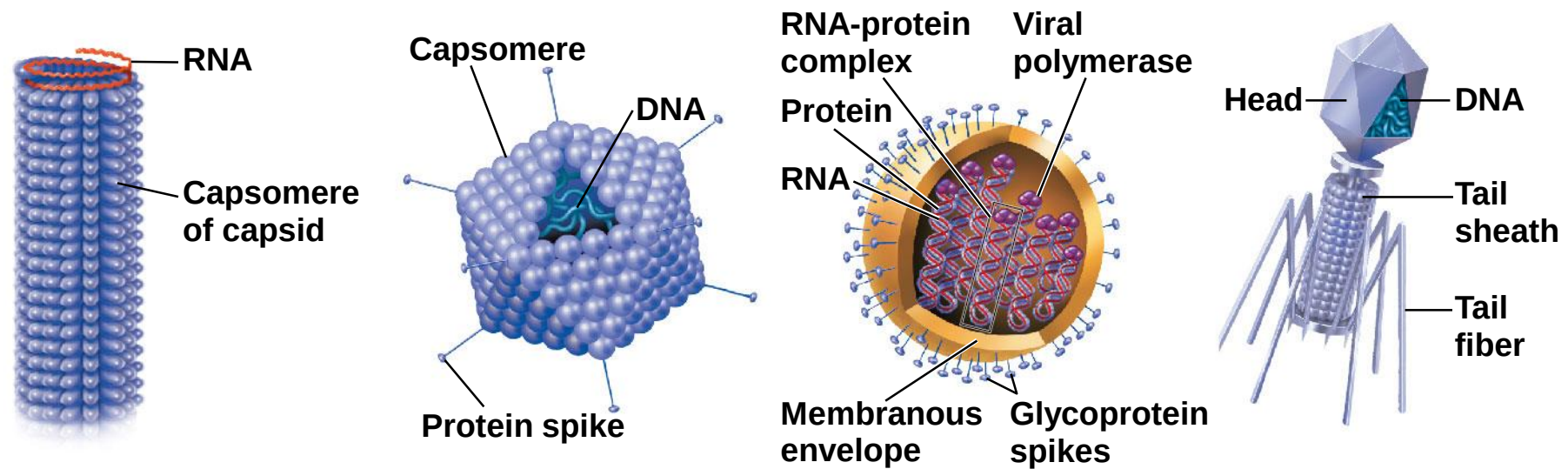
Viral Genomes

- Viral genomes may consist of either
 - double- or single-stranded DNA or
 - double- or single-stranded RNA
- Viruses are classified as DNA viruses or RNA viruses
- The genome is either a single linear or circular molecule of the nucleic acid
- Viruses have between three and 2,000 genes in their genome

Capsids and Envelopes

- A **capsid** is the protein shell that encloses the viral genome
- Capsids are built from protein subunits called capsomeres
- A capsid can have a variety of structures; associated viruses may be referred to as helical or icosahedral viruses

Figure 26.3

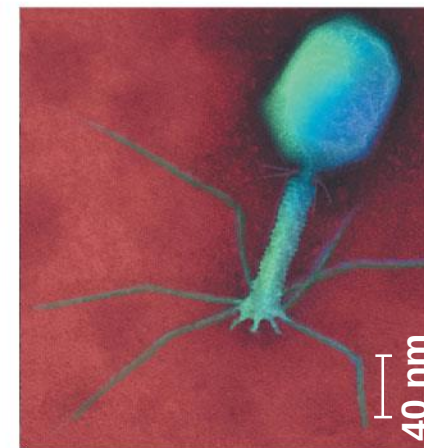
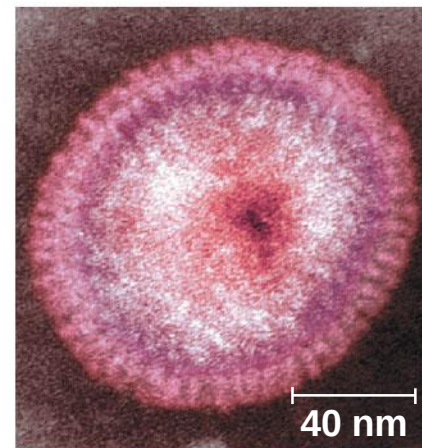
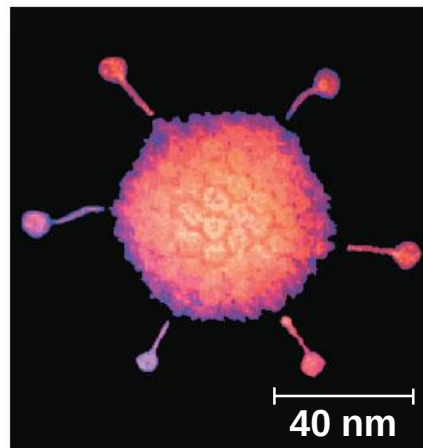
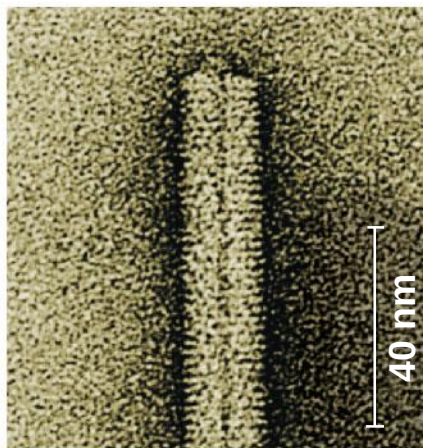


18 × 250 nm

70–90 nm (diameter)

80–200 nm (diameter)

80 × 225 nm



(a) Tobacco mosaic virus

(b) Adenoviruses

(c) Influenza viruses

(d) Bacteriophage T4

- Some viruses have accessory structures that help them infect hosts
- **Viral envelopes** (derived from membranes of host cells) surround the capsids of influenza viruses and many other viruses found in animals
- Viral envelopes contain a combination of viral and host cell molecules

- **Bacteriophages**, also called **phages**, are viruses that infect bacteria
- They have an elongated capsid head that encloses their DNA
- A protein tail piece attaches the phage to the host and injects the phage DNA inside

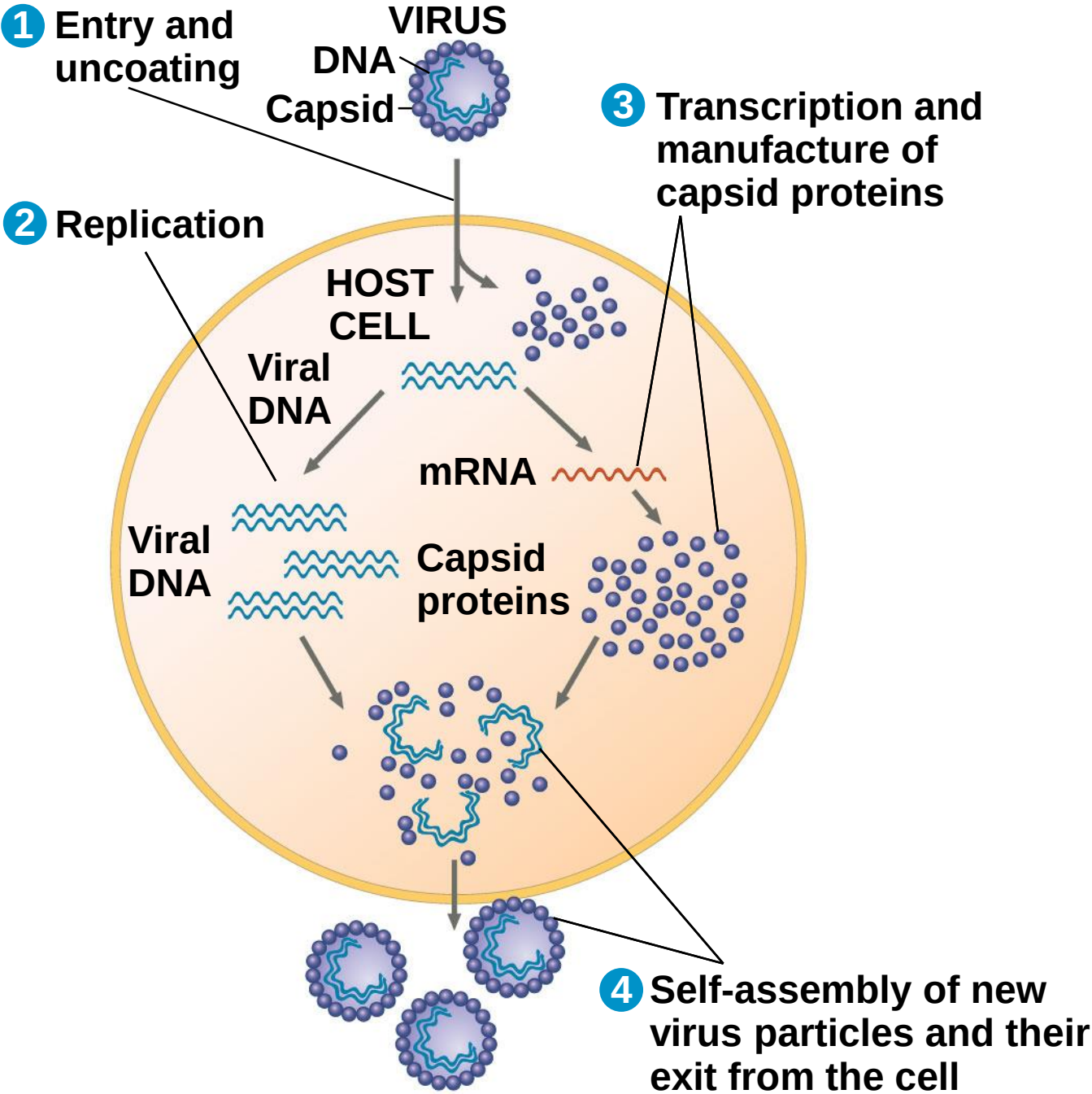
Concept 26.2: Viruses replicate only in host cells

- Viruses are obligate intracellular parasites, which means they can replicate only within a host cell
- Each virus has a **host range**, a limited number of host cells that it can infect

General Features of Viral Replicative Cycles

- The viral genome enters the host cell in a variety of ways
- Once a viral genome has entered a cell, the cell begins to manufacture viral proteins
- The virus makes use of host enzymes, ribosomes, tRNAs, amino acids, ATP, and other molecules
- Viral nucleic acid molecules and capsomeres spontaneously self-assemble into new viruses

Figure 26.4



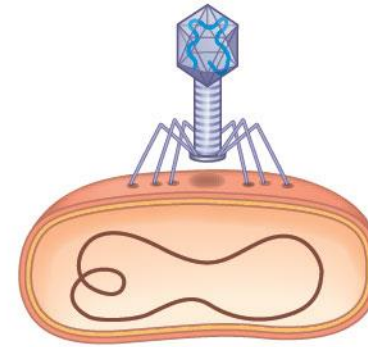
Replicative Cycles of Phages

- Phages are the best understood of all viruses
- Phages have two alternative reproductive mechanisms: the lytic cycle and the lysogenic cycle

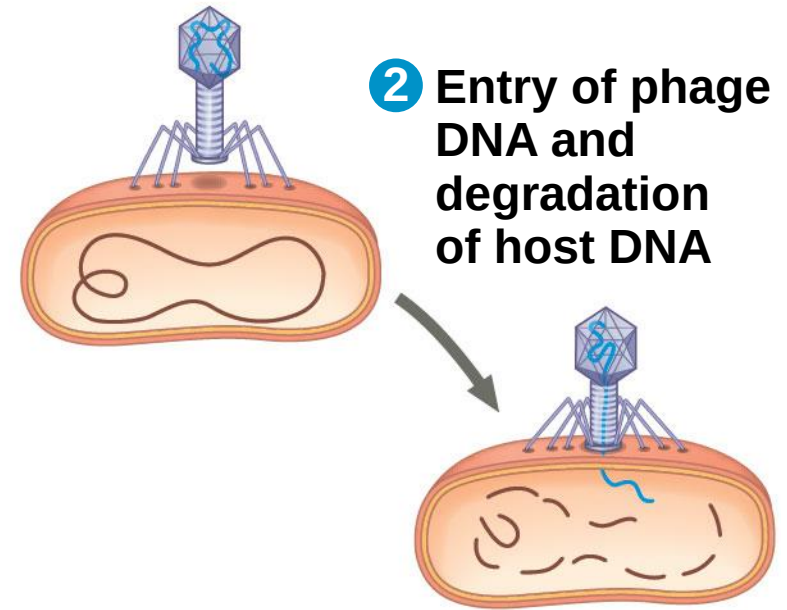
The Lytic Cycle

- The **lytic cycle** is a phage replicative cycle that culminates in the death of the host cell
- The lytic cycle produces new phages and lyses (breaks open) the host's cell wall, releasing the progeny viruses
- A phage that reproduces only by the lytic cycle is called a **virulent phage**

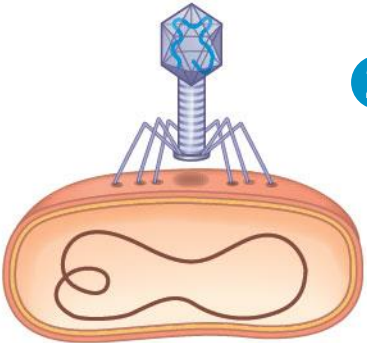
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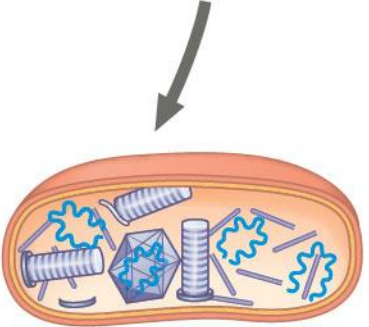
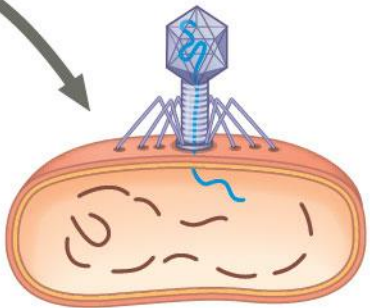
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2 Entry of phage DNA and degradation of host DNA



3 Synthesis of viral genomes and proteins

Figure 26.5_4

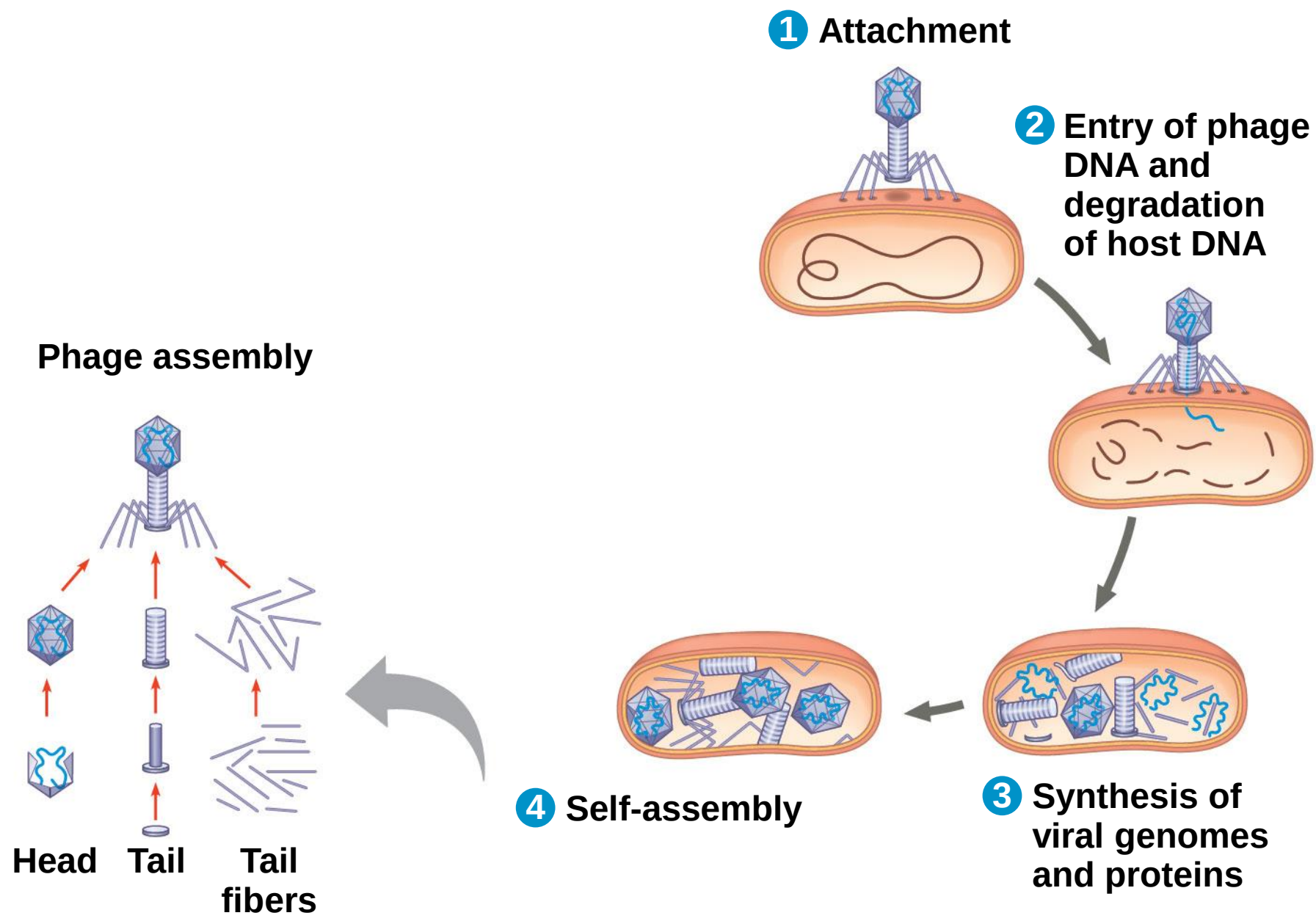
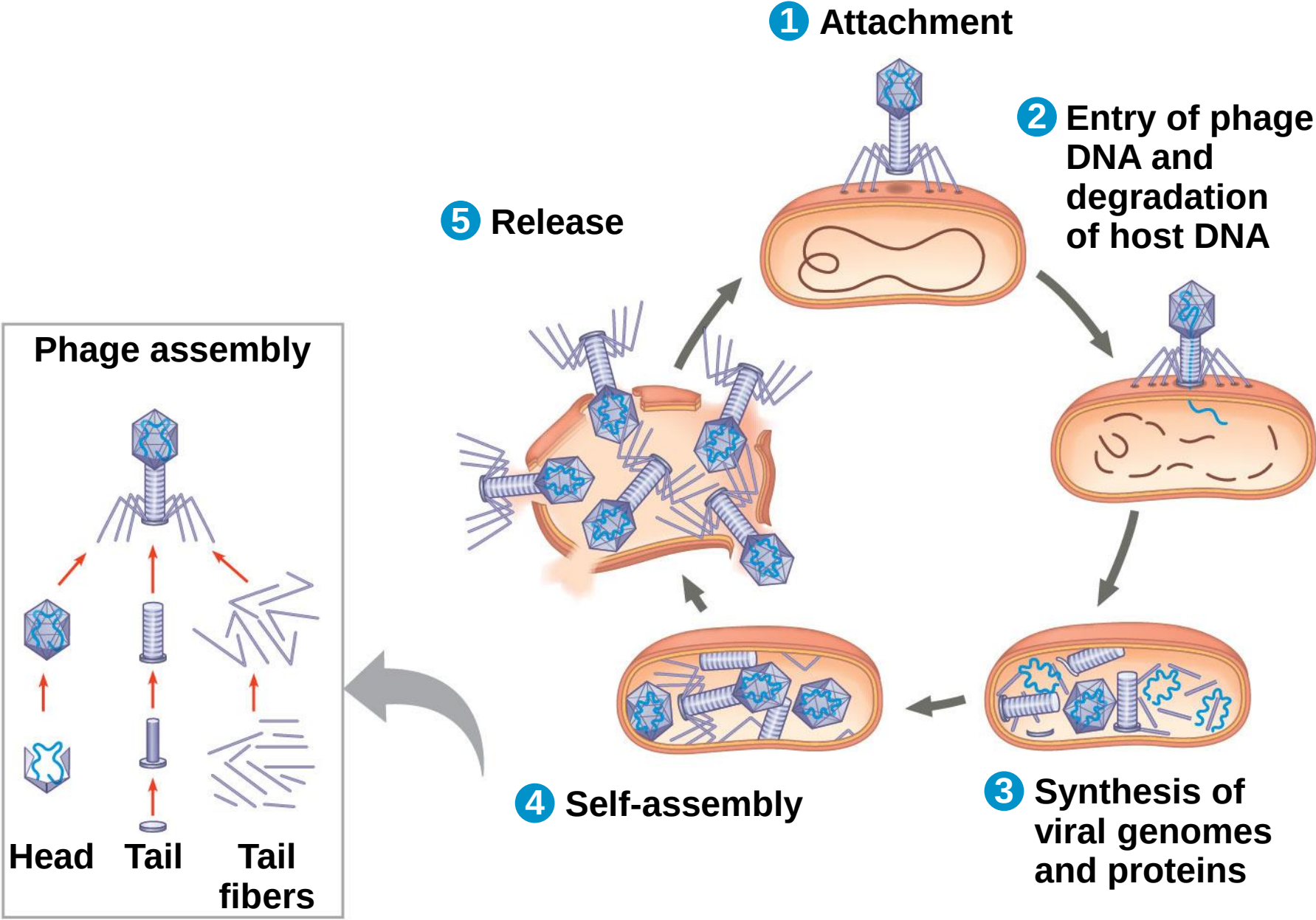


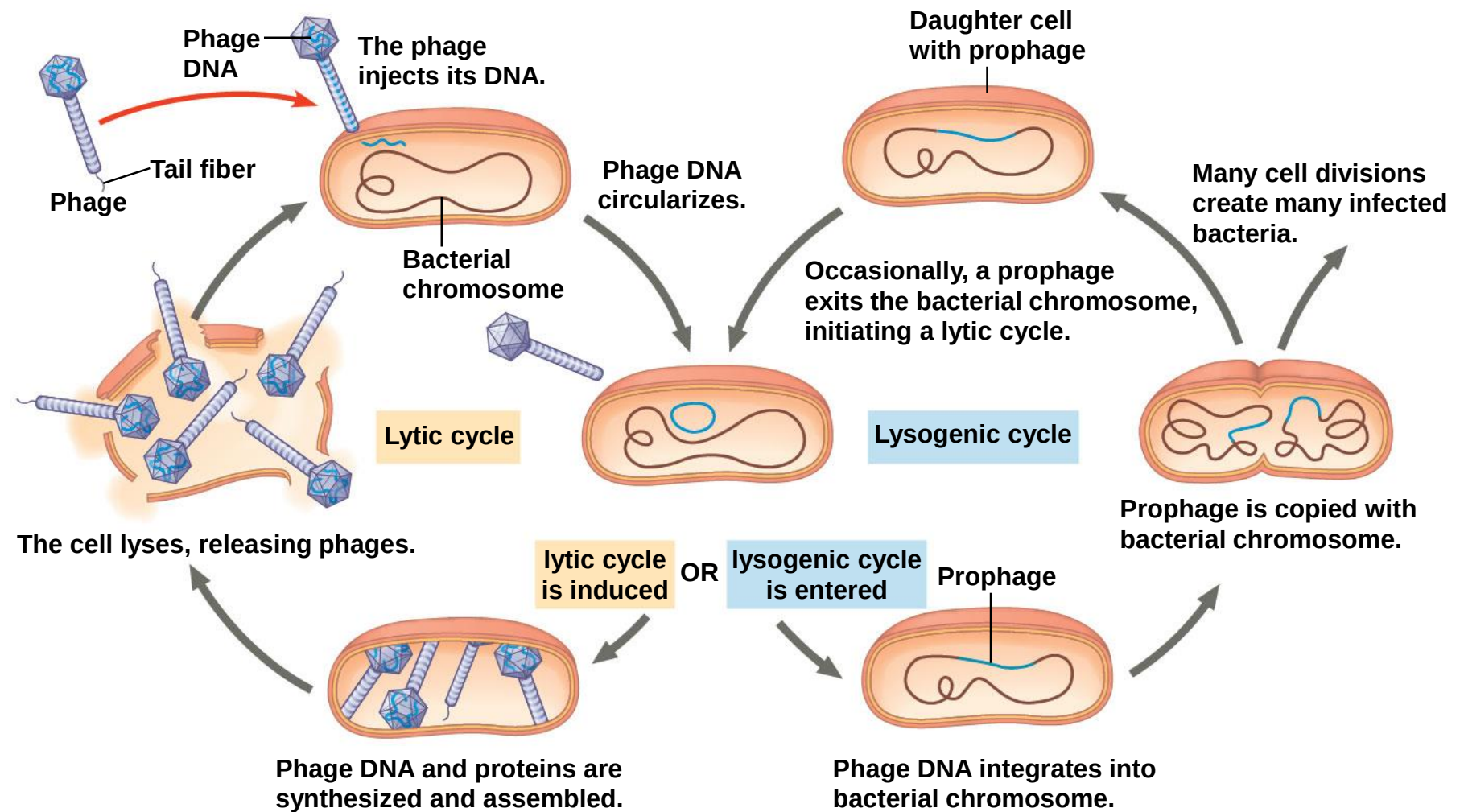
Figure 26.5_5



The Lysogenic Cycle

- The **lysogenic cycle** replicates the phage genome without destroying the host
- The viral DNA molecule is incorporated into the host cell's chromosome
- Phages that use both the lytic and lysogenic cycles are called **temperate phages**

Figure 26.6



- The integrated viral DNA is known as a **prophage**
- Every time the host divides, it copies the phage DNA and passes the copies to daughter cells
- An environmental signal can trigger the virus genome to exit the bacterial chromosome and switch to the lytic mode

Bacterial Defenses Against Phages

- Bacteria have their own defenses against phages
- Natural selection favors bacterial mutants with surface proteins that cannot be recognized as receptors by a particular type of phage
- Foreign DNA can be identified as such and cut up by cellular enzymes called **restriction enzymes**
- The bacterium's own DNA is protected from the restriction enzymes by being methylated

- Both bacteria and archaea can protect themselves from viral infection with the CRISPR-Cas system
- It is based on sequences called clustered regularly interspaced short palindromic repeats (CRISPRs)
- Each “spacer” sequence between the repeats corresponds to DNA from a phage that had infected the cell
- Particular nuclease proteins interact with the CRISPR region; these are called CRISPR-associated (Cas) proteins

- When a phage infects a bacterial cell that has the CRISPR-Cas system, the phage DNA is integrated between two repeat sequences
- If the cell survives the infection, it can block any attempt of the same type of phage to reinfect it
- The attempt of the phage to infect the cell triggers transcription of the CRISPR region
- The resulting RNAs are cut into pieces and bound by Cas proteins

- The Cas proteins use the phage-related RNA to target the invading phage DNA
- The phage DNA is cut and destroyed
- Natural selection favors phage mutants that can bind to altered cell surface receptors or that are resistant to enzymes
- The relationship between phage and bacteria is in constant evolutionary flux

Figure 26.7

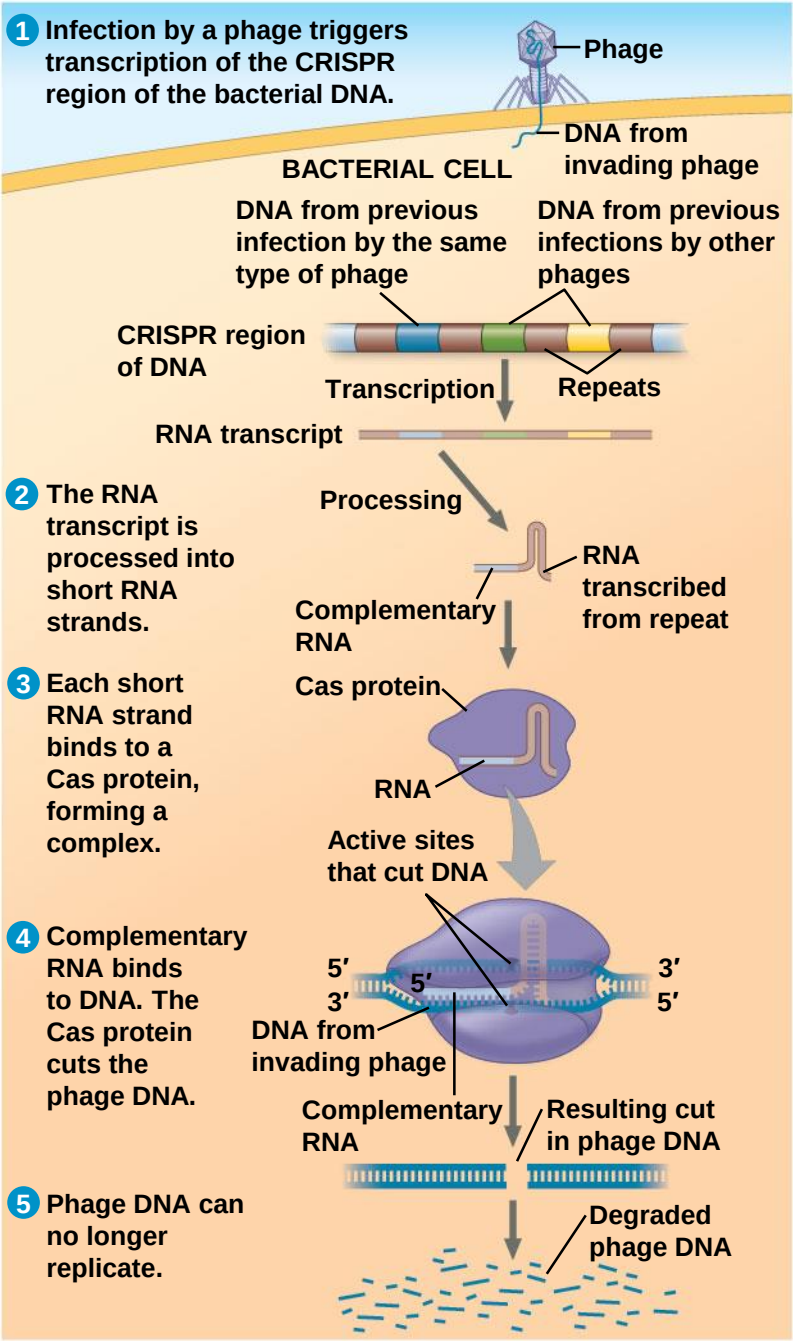


Figure 26.7a

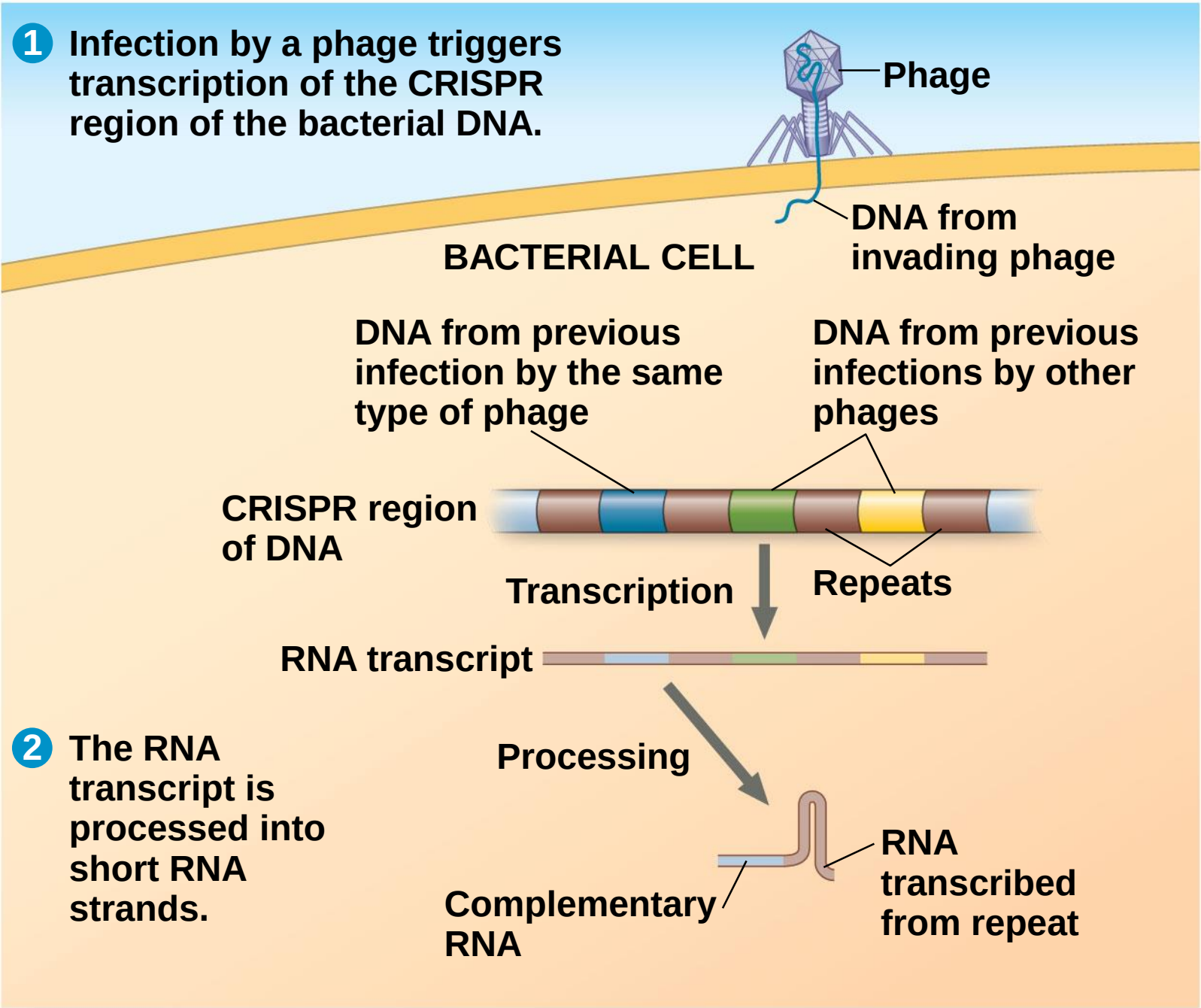
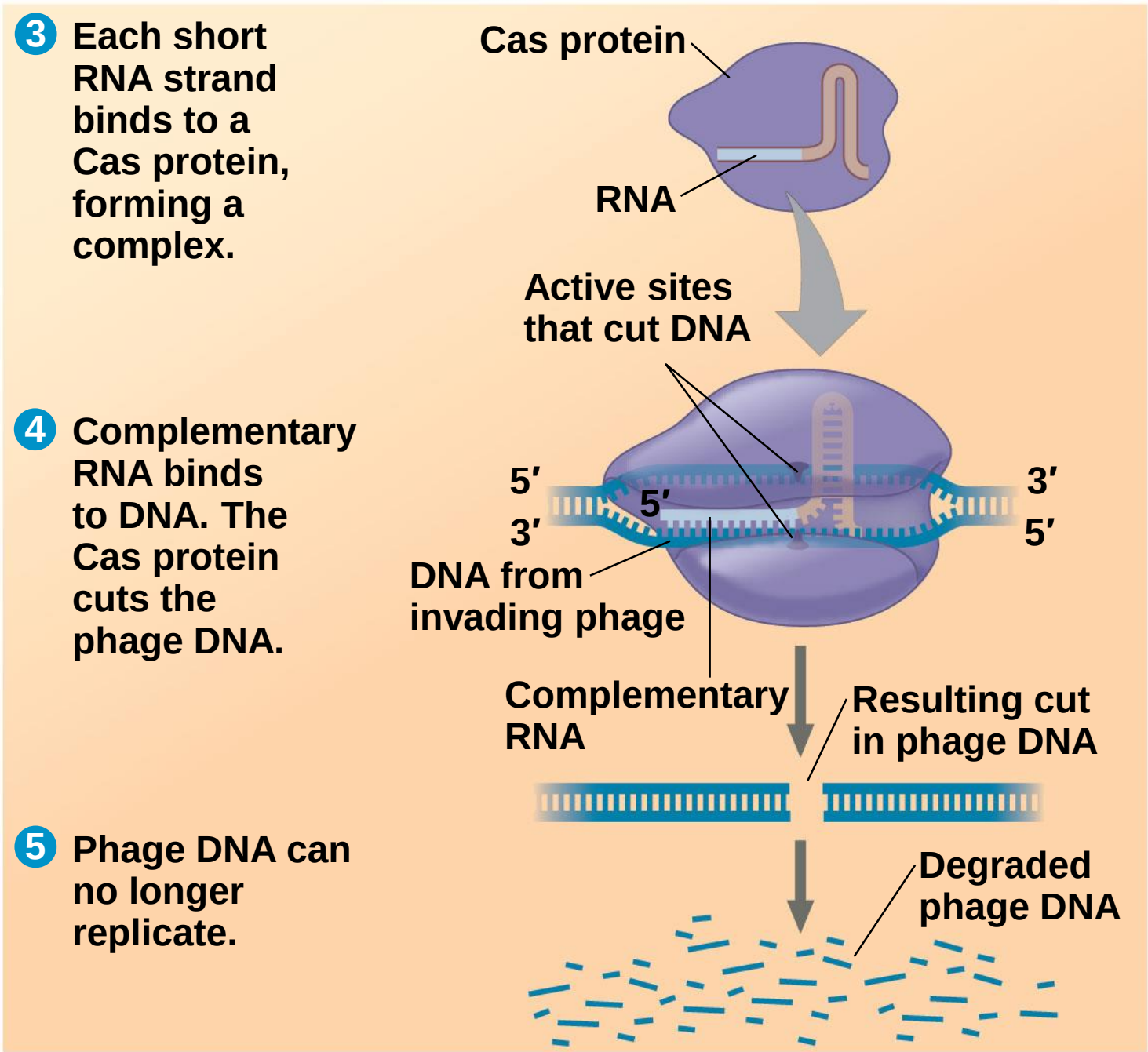


Figure 26.7b



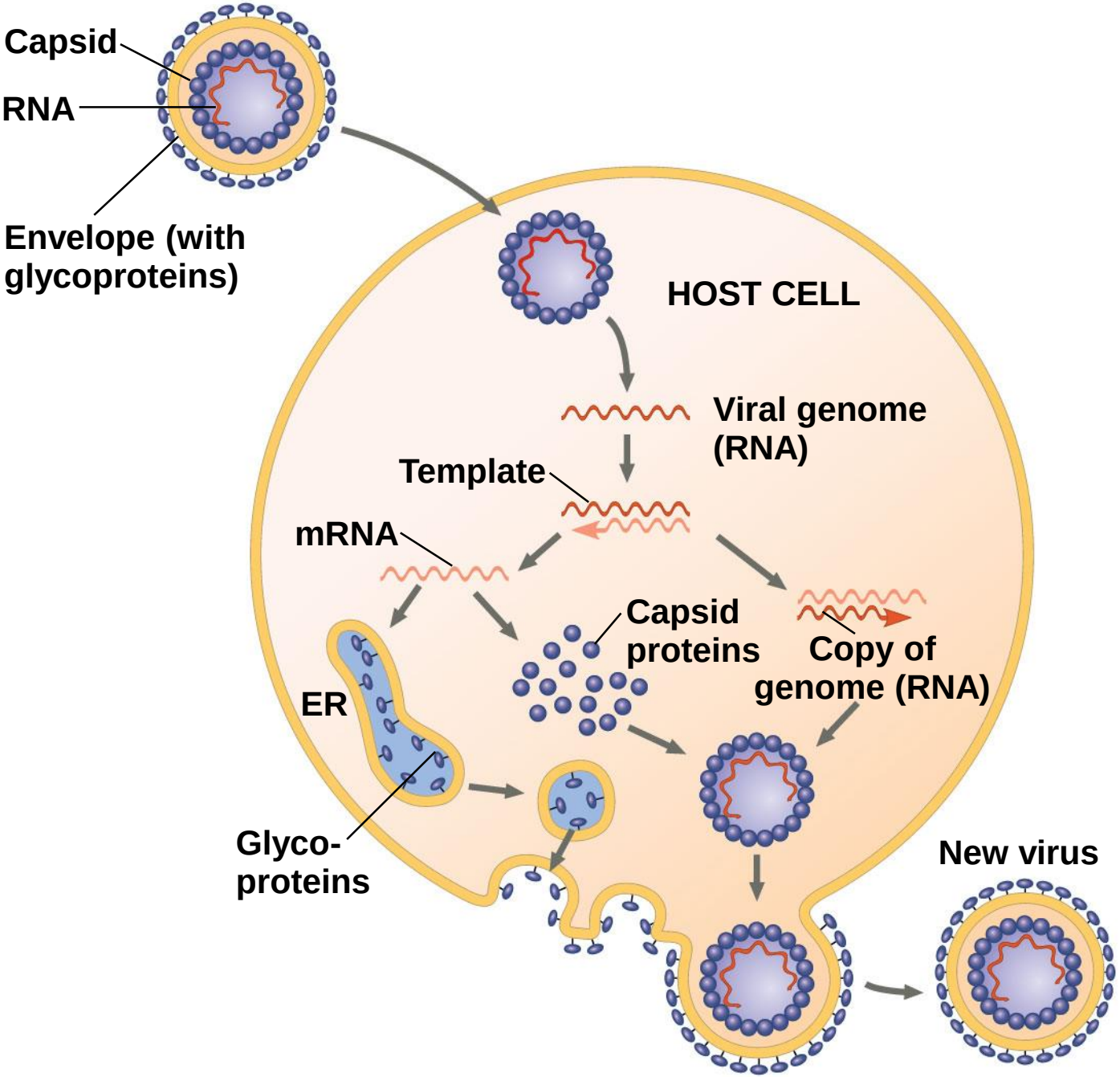
Replicative Cycles of Animal Viruses

- There are two key variables used to classify viruses that infect animals:
 - An RNA or DNA genome, either single-stranded or double-stranded
 - The presence or absence of a membranous envelope
- Whereas few bacteriophages have an envelope or an RNA genome, many animal viruses have both

Viral Envelopes

- Many viruses that infect animals have a membranous envelope
- Viral glycoproteins on the envelope bind to specific receptor molecules on the surface of a host cell
- The viral envelope is usually derived from the host cell's plasma membrane as the viral capsids exit

Figure 26.8



- Other viral membranes form from the host's nuclear envelope and are then replaced by an envelope made from Golgi apparatus membrane

Viral Genetic Material

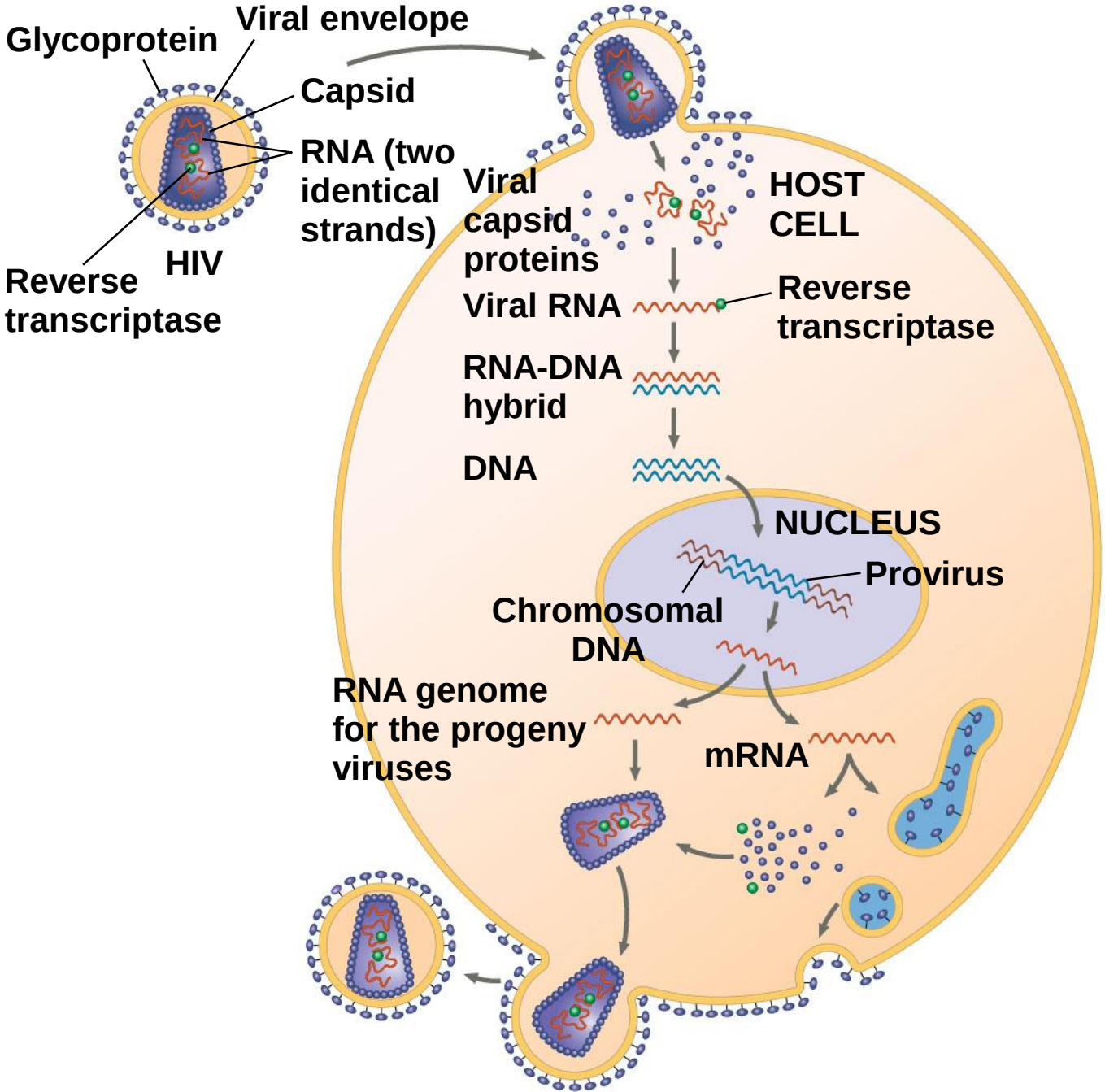
- The broadest variety of RNA genomes is found in viruses that infect animals
- **Retroviruses** use **reverse transcriptase** to copy their RNA genome into DNA
- **HIV (human immunodeficiency virus)** is the retrovirus that causes **AIDS (acquired immunodeficiency syndrome)**

- The viral DNA that is integrated into the host genome is called a **provirus**
- Unlike a prophage, a provirus remains a permanent resident of the host cell
- RNA polymerase transcribes the proviral DNA into RNA molecules
- The RNA molecules function both as mRNA for synthesis of viral proteins and as genomes for new virus particles released from the cell

Table 26.1

Table 26.1 Classes of Animal Viruses		
Class/Family	Envelope?	Examples That Cause Human Diseases
I. Double-Stranded DNA (dsDNA)		
Adenovirus (see Figure 19.3b)	No	Respiratory viruses
Papillomavirus	No	Warts, cervical cancer
Polyomavirus	No	Tumors
Herpesvirus	Yes	Herpes simplex I and II (cold sores, genital sores); varicella zoster (shingles, chicken pox); Epstein-Barr virus (mononucleosis, Burkitt's lymphoma)
Poxvirus	Yes	Smallpox virus; cowpox virus
II. Single-Stranded DNA (ssDNA)		
Parvovirus	No	B19 parvovirus (mild rash)
III. Double-Stranded RNA (dsRNA)		
Reovirus	No	Rotavirus (diarrhea); Colorado tick fever virus
IV. Single-Stranded RNA (ssRNA); Serves as mRNA		
Picornavirus	No	Rhinovirus (common cold); poliovirus; hepatitis A virus; other intestinal viruses
Coronavirus	Yes	Severe acute respiratory syndrome (SARS); Middle East respiratory syndrome (MERS)
Flavivirus	Yes	Zika virus (see Figure 19.10c); yellow fever virus; dengue virus; West Nile virus; hepatitis C virus
Togavirus	Yes	Chikungunya virus (see Figure 19.10b); rubella virus; equine encephalitis viruses
V. ssRNA; Serves as Template for mRNA Synthesis		
Filovirus	Yes	Ebola virus (hemorrhagic fever; see Figure 19.10a)
Orthomyxovirus	Yes	Influenza virus (see Figure 19.3c)
Paramyxovirus	Yes	Measles virus; mumps virus
Rhabdovirus	Yes	Rabies virus
VI. ssRNA; Serves as Template for DNA Synthesis		
Retrovirus	Yes	Human immunodeficiency virus (HIV/AIDS; see Figure 19.9); RNA tumor viruses (leukemia)

Figure 26.9



Evolution of Viruses

- Viruses do not fit our definition of living organisms
- Since viruses can replicate only within cells, they probably evolved as bits of cellular nucleic acid
- Candidates for the source of viral genomes include plasmids and transposons
- Plasmids, transposons, and viruses are all mobile genetic elements

- The largest virus yet discovered is the size of a small bacterium
- Its genome encodes proteins involved in translation, DNA repair, protein folding, and polysaccharide synthesis
- There is controversy about whether this virus evolved before or after cells

Concept 26.3: Viruses and prions are formidable pathogens in animals and plants

- Diseases caused by viral infections affect humans, agricultural crops, and livestock worldwide
- Smaller, less complex entities called prions also cause disease in plants and animals, respectively

Viral Diseases in Animals

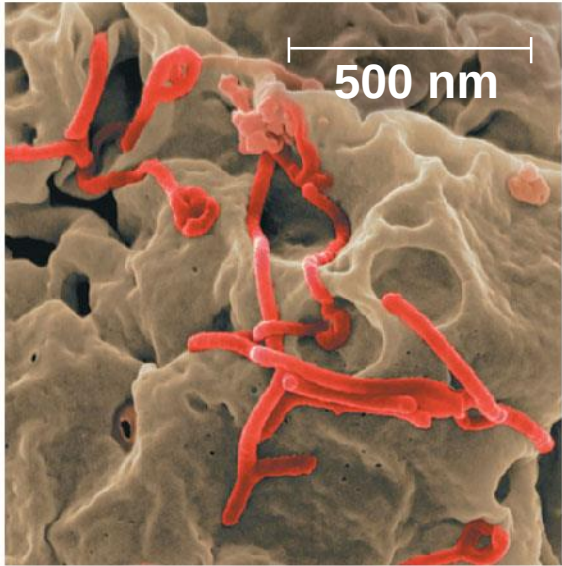
- Viruses may damage or kill cells by causing the release of hydrolytic enzymes from lysosomes
- Some viruses cause infected cells to produce toxins that lead to disease symptoms
- Others have molecular components such as envelope proteins that are toxic

- A **vaccine** is a harmless derivatives of pathogenic microbes that stimulate the immune system to mount defenses against the harmful pathogen
- Vaccines can prevent certain viral illnesses
- Viral infections cannot be treated by antibiotics
- Antiviral drugs can help to treat, not cure, viral infections by inhibiting synthesis of viral DNA and by interfering with viral assembly

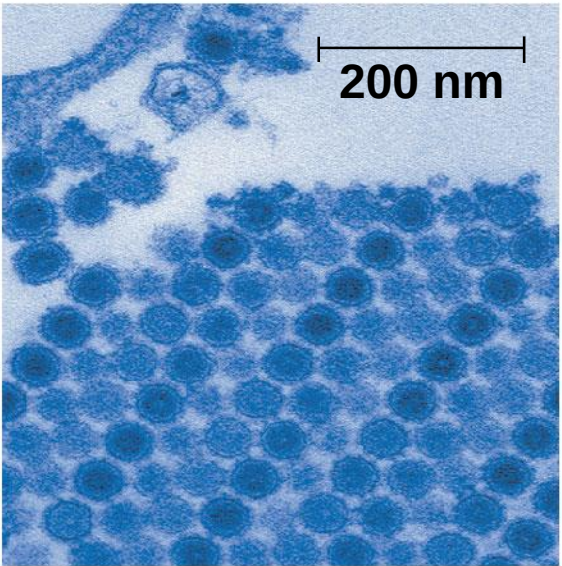
Emerging Viruses

- Emerging viruses are those that suddenly become apparent
- The Ebola virus is one of several emerging viruses that cause hemorrhagic fever, an often fatal illness
- Other examples include the chikungunya virus and the recently emerging Zika virus (2015)

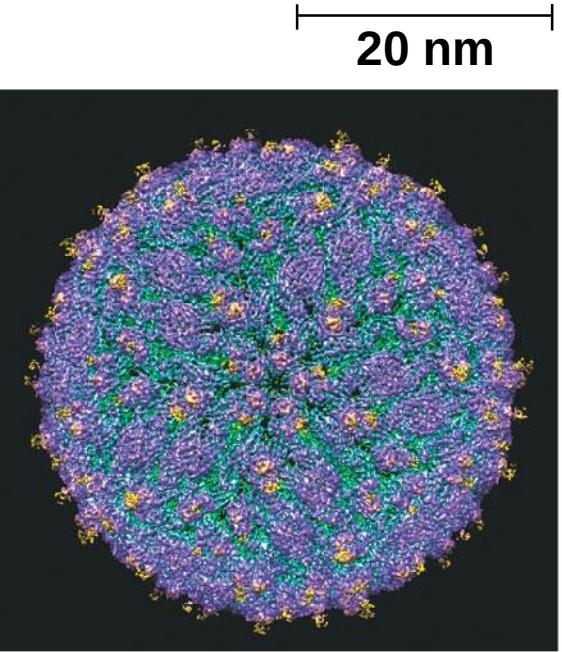
Figure 26.10



(a) Ebola viruses



(b) Chikungunya viruses



(c) Zika virus

- In 2009, a general outbreak (**epidemic**) of a flu-like illness appeared in Mexico and the United States, caused by an influenza virus named H1N1
- A global epidemic is called a **pandemic**

- Three processes contribute to the emergence of new viral diseases:
 - RNA viruses have an unusually high rate of mutation
 - The disease can be disseminated from a small, isolated human population and can eventually spread around the world
 - About three-quarters of new human diseases originate by spreading to humans from animals

- Flu epidemics are caused by type A influenza viruses; these infect a wide variety of animals including birds, pigs, horses, and humans
- Strains of influenza A are given standardized names based on the viral surface proteins hemagglutinin (HA) and neuraminidase (NA)
- H1N1 is the strain that caused the 2009 flu pandemic

- Changes in host behavior or the environment can increase the spread of viruses responsible for emerging diseases
- New roads into a remote area may increase spread of viral diseases
- The use of insecticides and mosquito nets may help prevent the spread
- It is possible that global climate change may allow mosquitoes that carry viruses to expand their range

Viral Diseases in Plants

- More than 2,000 types of viral diseases of plants are known and cause spots on leaves and fruits, stunted growth, and damaged flowers or roots
- Most plant viruses have an RNA genome
- Many have a helical capsid, while others have an icosahedral capsid

- Plant viruses spread disease by two major routes:
 - Horizontal transmission, entering through damaged cell walls
 - Vertical transmission, inheriting the virus from a parent

Prions: Proteins as Infectious Agents

- **Prions** are infectious proteins that appear to cause degenerative brain diseases in animals
- Scrapie in sheep, mad cow disease, and Creutzfeldt-Jakob disease in humans are all caused by prions
- Prions are incorrectly folded proteins, can be transmitted in food, act slowly, and are virtually indestructible

- Prions are somehow able to convert a normal form of the protein into the misfolded version
- Then several prions aggregate into a complex that can convert more proteins to prions
- Prions might also be involved in diseases such as Alzheimer's and Parkinson's disease

Figure 26.13

